

Progressive Resisted Exercises

(PRE)

- **Introduction**

- **PRE is a type of resistive exercises that is basically used to increase the strength of the patient's muscles.**
- **During the process, gradually the resistance is progressed (Increased) and hence it is named as "Progressive Resisted Exercises"**
- **It is a system of dynamic resistance training by some external load that provide resistance and it is further gradually increased.**
- **This technique was first described by "DeLorme and Watkins" in 1951**
- **DeLorme discussed about this technique in his book "PROGRESSIVE RESISTIVE EXERCISE :TECHNIC AND MEDICAL APPLICATION"**
- **This technique is widely used in Sports Physiotherapy especially in the athletes of weightlifting etc.**

- **In PRE, the contracting muscles are overloaded by a constant mechanical external load that is incrementally progressed.**
- **The basis of determining the resistance progression is “REPETITION MAXIMUM”**
- **Repetition Maximum is the greatest amount of weight a muscle can move through the Range of Motion a specific maximum no of time (repetition).**
 - **Eg: 5 kg 10 RM – Means that the person can lift the weight of 5kg maximum 10 times.**
- **There are different techniques of PRE.**

DeLorme Technique

- Classified as light to heavy approach
- Also called “DELORME-WATKINGS METHOD”
- **PROCEDURE**
 - First of all determine the 10 RM for the patient
 - Perform 3 sets of exercises as follows:
 1. 1st set – 10 repetition at 50% of 10 RM
 2. 2nd set – 10 repetition at 75% of 10 RM
 3. 3rd set – 10 repetition at 100% of 10 RM
 - For example, if the 10 RM of patient is 20 kg
 1. 1st set – 10 repetition at 10 kg
 2. 2nd set – 10 repetition at 15 kg
 3. 3rd set – 10 repetition at 20 kg

- **NOTE**

- **The patient perform all the three sets in each session daily or alternate.**
- **The patient can take a rest of few minutes between each sets.**
- **New 10 RM should be checked in each week and according to that new 10 RM, the progression should be made.**

Oxford Technique

- Classified as heavy to light approach
- Given by “Zinovieff”
- Reverse of Delorme
- **PROCEDURE**
 - First of all determine the 10 RM for the patient
 - Perform 3 sets of exercises as follows:
 1. 1st set – 10 repetition at 100% of 10 RM
 2. 2nd set – 10 repetition at 75% of 10 RM
 3. 3rd set – 10 repetition at 50% of 10 RM
 - For example, if the 10 RM of patient is 20 kg
 1. 1st set – 10 repetition at 20 kg
 2. 2nd set – 10 repetition at 15 kg
 3. 3rd set – 10 repetition at 10 kg

- **NOTE**

- **This technique was designed to be used in the beginning of rehabilitation and to diminish the resistance as muscle fatigue develop.**
- **This technique is used to decrease the effect of fatigue.**
- **Before starting this exercise, warm-up period of active exercises is needed.**

Oxford Technique

- The weight is neither reduced nor increased during the session
- Given in 4 sets each session
- **PROCEDURE**
 - First of all determine the 10 RM for the patient
 - Perform 4 sets of exercises as follows:
 1. 1st set – 10 repetition at 100% of 10 RM
 2. 2nd set – 10 repetition at 100% of 10 RM
 3. 3rd set – 10 repetition at 100% of 10 RM
 4. 4th set – 10 repetition at 100% of 10 RM
 - For example, if the 10 RM of patient is 20 kg
 1. 1st set – 10 repetition at 20 kg
 2. 2nd set – 10 repetition at 20 kg
 3. 3rd set – 10 repetition at 20 kg
 4. 4th set – 10 repetition at 100% of 10 RM

- **Precautions/Contraindication during PRE**
 - **Cardiovascular problems must be considered during exercises.**
 - **Avoid reaching to fatigue point**
 - **Give adequate time for recovery from fatigue**
 - **Avoid overwork to avoid overbreak of protein**
 - **Check for Osteoporosis to avoid fractures**
 - **Check for muscle soreness to avoid**